

Answer A

- 4) Using conservation of energy, the speed of the stone being thrown up will be equal to the speed of the stone travelling downwards at the same horizontal level.

So, the stone would be travelling downwards at 10 ms^{-1} from the edge of the cliff for 1.2 s with gravitational acceleration. Using kinematics equation $s = ut + \frac{1}{2}at^2$,
distance = $10(1.2) + 0.5(9.81)(1.2)^2 = 19.06 \text{ m}$

Answer D

- 5) For an elastic collision, the relative speed of approach = relative speed of separation